

Hamzah Ahmed

M.ENG. CANDIDATE / INTERNATIONAL ENERGY ENGINEERING

I build the models that decide how solar, storage and dispatch behave hour by hour, then prove in euros what the system earns. Before that I balanced air in live plant rooms, drew the pipes that carry it, and closed the contracts that paid for it.



CONTACT Amberg, Germany · +49 155 10603199 · h.ahmed.energy@gmail.com [PORTFOLIO →](#) [LINKEDIN →](#)

PERSONAL Bahraini, born 10 September 1997 · Student visa with Werkstudentenprivileg, up to 20 hours a week · Available now

SEEKING Werkstudent · internship · Master's thesis

🔌 BESS

Developed BESS investment software | Techno-economic modelling | Investment & dispatch | Python

☀️ Solar

Designed PV plant | 143 kWp rooftop PV in Polysun | CapEx & OpEx | LCOE | Yield simulation | Polysun

🌀 Wind

Modelled annual yield | Enercon E82 E2 yield model | Weibull distribution | Power curve analysis | MExcel

01 EDUCATION

Oct 2024
Oct 2027

International Energy Engineering, M.Eng.

OTH Amberg-Weiden, Germany

Energy Efficiency (1.0) · Wind & Hydropower (1.0) · Simulation of Energy Systems (1.3)
Energy Management with AI Methods (1.3) · Solar Energy (1.7) · Energy Law (1.7)

KEY PROJECTS

- BESS + PV investment software (Python, Streamlit)
- ORC, CCGT-CHP, boiler & heat pump optimization
- Microgrid simulation with CO₂ tracking
- 143 kWp rooftop PV design, NORMA GmbH
- Cradle-to-grave life cycle assessment, GaBi
- Enercon E82 E2 wind annual yield calculation

Sep 2015
Jun 2020

Mechanical Engineering, B.Sc.

University of Bahrain

Final project Gas-to-Power feasibility study – converting municipal bio-waste to biogas, then to electricity.

INTERNSHIPS

Jan – Feb 2020

Relay Engineering · Welding Inspection Engineer

Sep – Oct 2019

Bahrain Pharma · Maintenance Engineer

Jun – Aug 2019

Bahrain Steel · Engineering Consultant (Trainee)

02 EXPERIENCE

Sep 2022
Jul 2024

Key Account Lead, Aftersales

BFG International, Bahrain

- Grew yearly invoicing 12.5 % across Deutsche Bahn, SNCF, Talgo and Alstom.
- Answered and priced every technical enquiry from these rail clients.
- Redrew client design changes in CATIA and returned them to quoting.
- Built an Excel quoting tool and Power BI dashboard replacing manual reporting.
- Coordinated purchasing, logistics and engineering across warranty and retrofit work.

02 EXPERIENCE, CONTINUED

Apr 2022 Aug 2022	Testing and Balancing Engineer Commissioning Services International, Bahrain - Tested and balanced air and water HVAC systems on live building sites. - Recorded airflow, pressure and pump performance against the design spec.
Feb 2021 Aug 2021	Mechanical Engineer, Piping Networks TMAM Contracting, Bahrain - Drew piping and building services layouts in AutoCAD to international standards. - Prepared technical submittals and purchasing documents for approval.
Aug 2021 Apr 2022	<i>Between roles — self-taught Excel (VBA, macros) and Autodesk Inventor.</i>
Jul 2024 Sep 2024	<i>Relocation — Bahrain to Germany, enrolled at OTH Amberg-Weiden.</i>

03 TECHNICAL SKILLS

Energy Modelling	Python (oemof, Pyomo), LP and MILP optimisation, dispatch modelling, sector coupling
Storage & Systems	Battery storage (BESS), hydrogen and power-to-gas, microgrids, PV and wind yield
Data & Analytics	pandas, NumPy, SciPy, Power BI, Streamlit, time-series analysis
Engineering Tools	Excel (VBA, macros, CoolProp), MATLAB, CATIA, AutoCAD, Autodesk Inventor, Polysun
Life Cycle Analysis	GaBi advanced by Sphera — cradle-to-grave modelling
Certification	DataCamp — Data Analyst in Python (36 hours, April 2026)

04 LANGUAGES

English C1, IELTS certified
German B1, in daily use in Germany

05 OUTSIDE THE MODEL

Mixed martial arts · hiking · modern arts · professional street photography
A systems thinker by instinct — I turn optimisation models into interactive dashboards for the fun of it.

ACADEMIC REFERENCES	
Prof. Dr. Thomas Tiefel Technology and Innovation Management · OTH Amberg-Weiden +49 (0)9621 482-3324 · t.tiefel@oth-aw.de	Prof. Dr. Raphael Lechner Digital Energy Systems and Sector Coupling · OTH Amberg-Weiden +49 (0)9621 482-3330 · r.lechner@oth-aw.de

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Hamzah Ahmed
Amberg, October 8, 2026

Three years on real plant and real contracts, a master’s in energy engineering, and models that turn messy data into decisions worth defending.

Storage sizing · dispatch optimisation · energy system simulation. Available now — let’s talk.